Project Initialization and Planning Phase

Date	15 March 2026
Team ID	-
Project Title	Rising Waters - Flood Prediction Using Machine Learning
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) Report

This project develops a machine learning-based flood prediction system that analyzes rainfall, river level, weather, and historical flood data to predict flood risk. The solution helps disaster management authorities issue early warnings and reduce flood-related damage.

Objective	Develop an ML model for accurate flood prediction and early warning.
Scope	Predict flood risk using historical and real-time environmental data through a web application.
Problem Statement	Traditional forecasting methods are slower and less accurate for complex weather conditions.
Impact	Improves preparedness, reduces loss of life and property, and supports disaster management.
Approach	Train ML models using Kaggle datasets and deploy using Flask.
Key Features	• Flood risk prediction • Early warning alerts • Interactive dashboard • Historical analysis

Resource Requirements

Resource Type	Description	Specification/Allocation
Computing Resources	CPU	Intel i5/Ryzen 5
Memory	RAM	8 GB
Storage	Disk	256 GB SSD
Framework	Backend	Flask
Libraries	ML Libraries	Scikit-learn, Pandas, NumPy, Matplotlib
Development Environment	IDE	Visual Studio Code
Data	Dataset	Kaggle CSV Flood Dataset